

1、Naive Bayes 中 Gauss 模型的对数似然函数为：

$$\begin{aligned}
 L(\theta) = & C + \sum_{i=1}^n [y_i \ln p + (1-y_i) \ln(1-p)] \\
 & + \sum_{i=1}^n y_i \left[-\frac{1}{2} \ln \det(\Sigma_1) - \frac{1}{2} (\mathbf{x}_i - \boldsymbol{\mu}_1)^T \Sigma_1^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_1) \right] \\
 & + \sum_{i=1}^n (1-y_i) \left[-\frac{1}{2} \ln \det(\Sigma_0) - \frac{1}{2} (\mathbf{x}_i - \boldsymbol{\mu}_0)^T \Sigma_0^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_0) \right]
 \end{aligned}$$

其中 C 为任意常数，求 $\theta^* = \arg \max L(\theta)$ ，其中 $\theta = \{p, \boldsymbol{\mu}_1, \Sigma_1, \boldsymbol{\mu}_0, \Sigma_0\}$ 。

解：

$$(1) \quad \frac{\partial L}{\partial p} = \sum_{i=1}^n \left[(y_i) \frac{\partial}{\partial p} \ln p + (1-y_i) \frac{\partial}{\partial p} \ln(1-p) \right] = \sum_{i=1}^n \left(\frac{y_i}{p} - \frac{1-y_i}{1-p} \right) = 0,$$

$$\text{两边同乘 } np(1-p) \text{ 得： } \sum_{i=1}^n [(1-p)y_i - p(1-y_i)] = \sum_{i=1}^n (y_i - p) = 0,$$

$$\text{解得： } p = \frac{1}{n} \sum_{i=1}^n y_i.$$

(2)

$$\frac{\partial L}{\partial \boldsymbol{\mu}_1} = \sum_{i=1}^n y_i \left[-\frac{1}{2} \frac{\partial}{\partial \boldsymbol{\mu}_1} (\mathbf{x}_i - \boldsymbol{\mu}_1)^T \Sigma_1^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_1) \right] = \sum_{i=1}^n y_i \left[\frac{1}{2} (\Sigma_1^{-1} + \Sigma_1^{-T}) (\mathbf{x}_i - \boldsymbol{\mu}_1) \right] = 0,$$

由于 Σ_1 是对称矩阵，故整理得： $\sum_{i=1}^n y_i \Sigma_1^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_1) = 0$ ，两边同乘 $n\Sigma_1$ 得：

$$\sum_{i=1}^n y_i (\mathbf{x}_i - \boldsymbol{\mu}_1) = 0, \quad \text{即 } \sum_{i=1}^n y_i \mathbf{x}_i = \boldsymbol{\mu}_1 \sum_{i=1}^n y_i, \quad \text{解得： } \boldsymbol{\mu}_1 = \frac{\sum_{i=1}^n y_i \mathbf{x}_i}{\sum_{i=1}^n y_i}.$$

同理可得： $\mu_0 = \frac{\sum_{i=1}^n (1-y_i) \mathbf{x}_i}{\sum_{i=1}^n (1-y_i)}$ 。

(3) 令 $\Lambda = \Sigma_1^{-1}$ ，则：

$$\begin{aligned} \frac{\partial}{\partial \Lambda} &= \sum_{i=1}^n y_i \left[-\frac{1}{2} \frac{\partial}{\partial \Lambda} \ln \det(\Lambda^{-1}) - \frac{1}{2} \frac{\partial}{\partial \Lambda} (\mathbf{x}_i - \mu_1)^T \Lambda (\mathbf{x}_i - \mu_1) \right] \\ &= \sum_{i=1}^n \frac{y_i}{2} \left[\frac{\partial}{\partial \Lambda} \ln \det(\Lambda) - (\mathbf{x}_i - \mu_1)(\mathbf{x}_i - \mu_1)^T \right] \\ &= \sum_{i=1}^n \frac{y_i}{2} [\Lambda^{-T} - (\mathbf{x}_i - \mu_1)(\mathbf{x}_i - \mu_1)^T] = 0 \end{aligned}$$

由于 Σ_1 是可逆、对称矩阵，故 $\Lambda^{-T} = \Sigma_1^T = \Sigma_1$ ，于是：

$$\sum_{i=1}^n y_i \Sigma_1 = \sum_{i=1}^n y_i (\mathbf{x}_i - \mu_1)(\mathbf{x}_i - \mu_1)^T, \text{ 解得: } \Sigma_1 = \frac{\sum_{i=1}^n y_i (\mathbf{x}_i - \mu_1)(\mathbf{x}_i - \mu_1)^T}{\sum_{i=1}^n y_i}。$$

$$\text{同理可得: } \Sigma_0 = \frac{\sum_{i=1}^n (1-y_i)(\mathbf{x}_i - \mu_0)(\mathbf{x}_i - \mu_0)^T}{\sum_{i=1}^n (1-y_i)}。$$

此处用了一些矩阵与矩阵微分的结论：

(1) $\det(\Lambda^{-1}) = \det(\Lambda)^{-1}$ ，矩阵与其逆的行列式的值互为倒数；

(2) $\frac{\partial}{\partial A} \ln \det(A) = A^{-T}$ ；

(3) $\mathbf{x}^T A \mathbf{x} = \text{tr}(\mathbf{x}^T A \mathbf{x}) = \text{tr}(\mathbf{x} \mathbf{x}^T A) = \text{tr}(A \mathbf{x} \mathbf{x}^T)$ ；

$$(4) \quad \frac{\partial}{\partial A} \text{tr}(BA) = B^T。$$

2、Naive Bayes 中 Bernoulli 模型的对数似然函数为：

$$\begin{aligned} L(\theta) = & \sum_{i=1}^n y_i \ln \varphi_1 + (1 - y_i) \ln(1 - \varphi_1) + \\ & \sum_{i=1}^n y_i \sum_{j=1}^d [x_{ij} \ln \varphi_{j1} + (1 - x_{ij}) \ln(1 - \varphi_{j1})] + \\ & \sum_{i=1}^n (1 - y_i) \sum_{j=1}^d [x_{ij} \ln \varphi_{j0} + (1 - x_{ij}) \ln(1 - \varphi_{j0})] \end{aligned}$$

求 $\theta^* = \arg \max L(\theta)$ ，其中 $\theta = \{\varphi_1, \varphi_{j1}, \varphi_{j0}\}$ ， $j = 1, 2, \dots, d$ 。

解：

$$(1) \quad \frac{\partial L}{\partial \varphi_1} = \sum_{i=1}^n \left(\frac{y_i}{\varphi_1} - \frac{1 - y_i}{1 - \varphi_1} \right) = 0，\text{由第一题的结果易得：} \varphi_1 = \frac{1}{n} \sum_{i=1}^n y_i。$$

$$(2) \quad \frac{\partial L}{\partial \varphi_{j1}} = \sum_{i=1}^n y_i \left(\frac{x_{ij}}{\varphi_{j1}} - \frac{1 - x_{ij}}{1 - \varphi_{j1}} \right) = 0，\text{解得：} \varphi_{j1} = \frac{\sum_{i=1}^n y_i x_{ij}}{\sum_{i=1}^n y_i}。$$

$$(3) \quad \text{同理得：} \varphi_{j0} = \frac{\sum_{i=1}^n (1 - y_i) x_{ij}}{\sum_{i=1}^n (1 - y_i)}。 \text{在(2)(3)中 } j = 1, 2, \dots, d。$$